

## Part-C Case Study

### Case Study 1 – Student Performance Analytics

You are given data of 100 students containing their marks in Math, Science, and English.

- (a) Using Pandas, calculate the average marks for each student and identify the top 5 performers.
- (b) Compute subject-wise average and standard deviation; comment which subject shows the highest variation.
- (c) Draw a boxplot for each subject and comment on outliers.

```
In [1]: # 1(a)
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV
df=pd.read_csv("Data/Students_Marks.csv")

# Calculate average marks
df['Average']=df[['Math','Science','English']].mean(axis=1)

# Top 5 performers
top=df.sort_values(by='Average',ascending=False).head(5)
print("Top Five Performers: \n",top[['Math','Science','English','Average']])
```

Top Five Performers:

|    | Math | Science | English | Average   |
|----|------|---------|---------|-----------|
| 65 | 92   | 96      | 92      | 93.333333 |
| 0  | 78   | 96      | 99      | 91.000000 |
| 59 | 99   | 91      | 76      | 88.666667 |
| 33 | 83   | 99      | 84      | 88.666667 |
| 44 | 90   | 97      | 74      | 87.000000 |

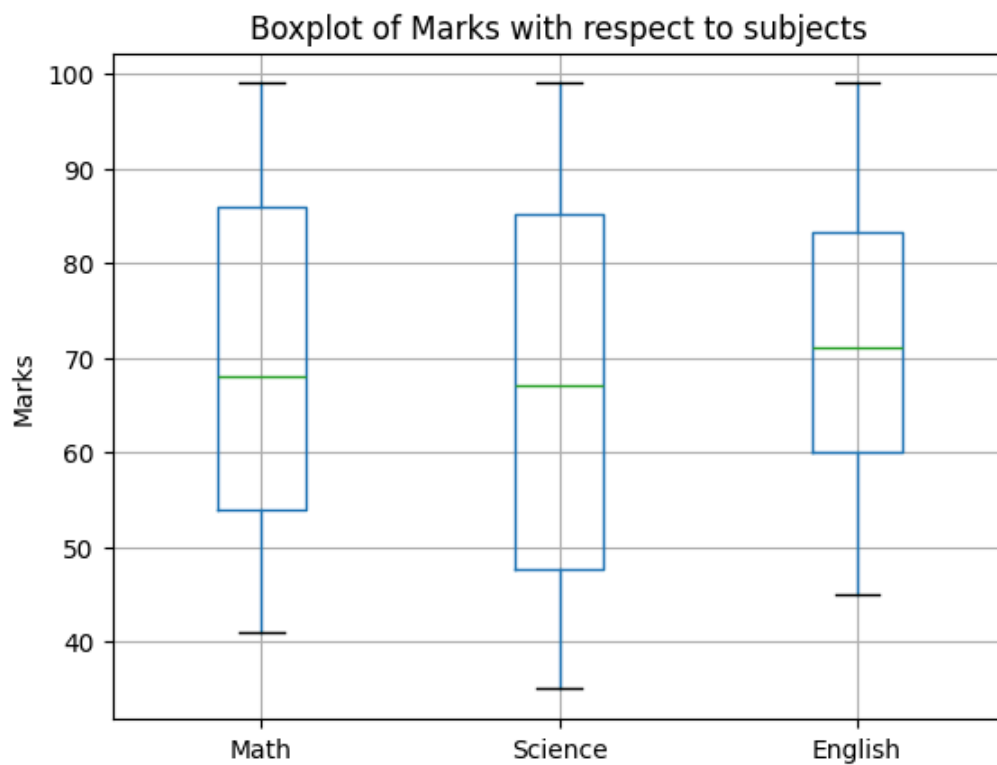
```
In [2]: # 1(b)
subject=df[['Math','Science','English']].agg(['mean','std'])
print("\nSubject-wise Average and Standard Deviation: \n",subject)
var=subject.loc['std'].idxmax()
print(f"\nSubject with highest variance: {var}")
```

Subject-wise Average and Standard Deviation:

|      | Math      | Science   | English   |
|------|-----------|-----------|-----------|
| mean | 69.580000 | 66.880000 | 71.320000 |
| std  | 18.031499 | 20.386805 | 15.868067 |

Subject with highest variance: Science

```
In [3]: # 1(c)
df[['Math','Science','English']].boxplot()
plt.title("Boxplot of Marks with respect to subjects")
plt.ylabel("Marks")
plt.show()
```



```
In [4]: # Outliers are present mainly in the lower ranges of Science and Math marks (<40),
# indicating a few students performed poorly in these subjects.
# English shows fewer outliers and less spread, meaning performance is more consistent.
# Science has the highest variation overall.
```

## Case Study 2 – Sales Data Exploration

A company records sales transactions in a dataset with columns: Product, Quantity, Price, Region.

- (a) Add a new column Total = Quantity × Price and compute overall revenue.
- (b) Group data by region and identify which region contributes the most sales.
- (c) Plot a histogram of total sales per product and discuss which products are underperforming.

```
In [5]: # 2(a)
import pandas as pd
import matplotlib.pyplot as plt

# Read CSV
df=pd.read_csv("Data/Sales_Data.csv")

# Add Total column and compute overall revenue
df['Total']=df['Quantity']*df['Price']
revenue=df['Total'].sum()
print("Overall Revenue:",revenue)
```

Overall Revenue: 1177230.1700000004

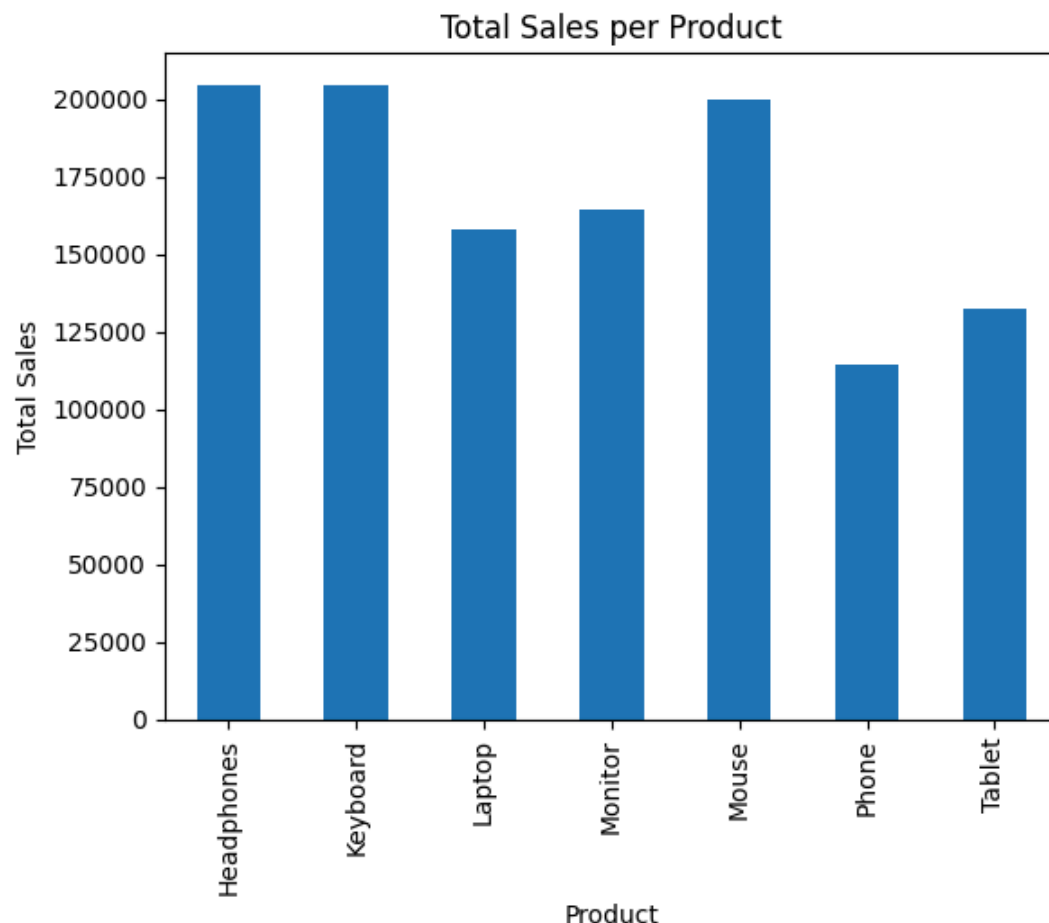
```
In [6]: # 2(b)
region=df.groupby('Region')['Total'].sum().sort_values(ascending=False)
print("\nSales by Region: \n",region)
top=region.idxmax()
print(f"\nRegion with highest sales: {top}")
```

Sales by Region:

```
Region
South    369727.81
North    286862.46
East     274176.60
West     246463.30
Name: Total, dtype: float64
```

Region with highest sales: South

```
In [7]: # 2(c)
product=df.groupby('Product')['Total'].sum()
product.plot(kind='bar',title='Total Sales per Product')
plt.ylabel('Total Sales')
plt.xlabel('Product')
plt.show()
```



```
In [8]: # By the help of data visualization from the above bar graph we can simply state that
# Product - Phone is underperforming as the total sales is least when compared with other produc
```

### Case Study 3 – Predicting Fuel Efficiency (Auto MPG Dataset)

You are analyzing the Auto MPG dataset (UCI Repository) with features: mpg, cylinders, displacement, horsepower, weight, acceleration, model year, origin.

- (a) Compute the correlation of mpg with each numeric feature and discuss which factors influence fuel efficiency most.
- (b) Plot a scatter plot of weight vs mpg; describe the trend you observe.

```
In [9]: # 3(a)
import pandas as pd
import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings("ignore", category=FutureWarning)
```

```
col=["mpg","cylinders","displacement","horsepower","weight",
      "acceleration","model_year","origin","car_name"]

# Adding the url to the data sets
df=pd.read_csv("https://archive.ics.uci.edu/ml/machine-learning-databases/auto-mpg/auto-mpg.data",
               names=col,delim_whitespace=True,na_values='?')

# Drop rows with missing horsepower or convert appropriately
df=df.dropna(subset=['horsepower'])
df['horsepower']=df['horsepower'].astype(float)

# Compute correlations of mpg with each numeric feature
corrs=df[['mpg','cylinders','displacement','horsepower',
          'weight','acceleration','model_year','origin']].corr()['mpg'].drop('mpg')
print("Correlation of mpg with other features:\n",corrs.sort_values(ascending=False))
```

Correlation of mpg with other features:

```
model_year    0.580541
origin        0.565209
acceleration  0.423329
cylinders     -0.777618
horsepower    -0.778427
displacement  -0.805127
weight        -0.832244
Name: mpg, dtype: float64
```

In [10]: *# mpg shows strong negative correlation with weight, displacement, and horsepower, meaning heavier cars with larger engines tend to have lower fuel efficiency. Model year has a positive correlation with mpg, indicating newer models are more efficient. Cylinders also correlate negatively, as more cylinders generally reduce mpg. Hence, vehicle weight and engine power are the main factors influencing fuel efficiency.*

In [11]: *# 3(b)*  
plt.scatter(df['weight'],df['mpg'],alpha=0.6)  
plt.xlabel("Weight")  
plt.ylabel("MPG")  
plt.title("Scatter: Weight vs MPG")  
plt.show()

